

Frontier Models for Quant Research & AI Engineering

MiniMax-M3 · Kimi K3 · Claude Opus 4.8 — with Fable 5, GPT-5.6 Sol, and Gemini 3 Pro as reference points

Six frontier-class models, one question: which earns the default seat in a quant-research and AI-engineering workflow — and what should route where. A metrics-first comparison, with every number sourced and every gap declared.

Data as of 18 July 2026 (Kimi K3 released 16 July 2026)

Prepared by Christian Macion · STELLA Research

Method Multi-agent research pipeline with two independent verification gates

Distribution Public — sources footnoted throughout

Contents

- 1. Executive summary
- 2. The landscape
- 3. Metrics for quant research
- 4. Metrics for AI engineering & architecture
- 5. Operational metrics
- 6. Cost-per-task economics
- 7. The macro frame: why the price structure looks like this
- 8. Decision framework
- 9. Method, verification & gap log

1. Executive summary

**10–
12.5×**

MiniMax-M3 is cheaper per token than Kimi K3 — the market floor among frontier-class APIs

88.3

Kimi K3 Terminal-Bench 2.1 — above Opus 4.8 (84.6), within 0.5 of GPT-5.6 Sol (88.8)

f ≈ 44%

Cache-hit fraction where Opus 4.8's effective input price undercuts K3's base input price

**144.9
t/s**

M3 measured output throughput (independent) — ~5× K3's ~30 t/s community reports

The landscape rearranged itself in 48 hours. On 16 July 2026 Moonshot shipped Kimi K3 — a 2.8-trillion-parameter sparse MoE model at \$3/\$15 per million tokens — and within a day published agentic benchmarks that place it in the frontier cluster: Terminal-Bench 2.1 **88.3** (above Opus 4.8's 84.6, just under GPT-5.6 Sol's 88.8), MCP Atlas **84.2**, and BrowseComp **91.2**.¹ The release-day claim that K3 “has no agentic numbers” was true for roughly eleven hours. It is no longer a usable premise.

The quieter finding is that MiniMax-M3 — the incumbent in this evaluation — is the market floor. M3's public pricing, confirmed against MiniMax's platform documentation and three corroborating sources, is **\$0.30/\$1.20 per million tokens** with a 1M-token context window.² That is 10–12.5× cheaper than K3 and roughly 17–21× cheaper than Opus 4.8 standard. M3 is not a weak model: its vendor-published SWE-bench Verified is **80.5** (between Gemini 3 Pro's 76.2 and Opus 4.8's 88.6), and its independently measured throughput of **~120–145 tokens/second** is the fastest in this comparison by a wide margin.³

Capability per dollar now has three distinct tiers. At the top, Claude Fable 5 (\$10/\$50) leads general reasoning (HLE 53.3, SWE-bench 95.0) and commands the highest price. In the middle, K3, Opus 4.8, GPT-5.6 Sol, and Gemini 3 Pro trade blows within a few points on most benchmarks at \$2–\$30. At the bottom, M3 delivers 90–95% of the middle tier’s measured capability at one-tenth the price. The correct question is not “which model is best” but “**which tier does each workload deserve.**”

Cache economics complicate the middle. Opus 4.8’s prompt-cache minimum dropped to 1,024 tokens (from 4,096), which means most real agent loops — with multi-kilobyte static system prompts — now qualify for a 90% input discount.⁴ Derivation in §6: above a ~**44% cache-hit fraction**, Opus 4.8’s effective input price falls below K3’s \$3 base. Agentic workloads with 70–90% repeated prefixes can therefore be *cheaper on Opus than on K3*, despite K3’s lower sticker — while K3’s own cache price (\$0.30) is published without documented mechanics. K3 also runs a single “max” reasoning tier, and verbosity is real: one independent test drew 16,658 output tokens (13,241 reasoning) from a 95-token prompt.⁵

For quant research specifically, the benchmark picture is unusually tight. GPQA-Diamond separates the six models by only 2.2 points (91.9–94.1). The differentiators are instead: long-horizon synthesis quality (K3 leads Artificial Analysis’s long-horizon knowledge-work Elo among non-Fable models), finance-numeric benchmarks (none published for any model — an honest null), and hallucination discipline on citation-heavy work (Anthropic self-reports the lowest incorrect-rate among six tested models for Opus 4.8).

The recommendation in one paragraph: run a **three-tier routing policy** — M3 for bulk and high-frequency work (its price is a structural advantage, not a rounding error), K3 for long-horizon synthesis and terminal-agentic tasks where its numbers genuinely lead, Opus 4.8 for cache-friendly agent loops and precision-critical writing — with Fable 5 reserved for the small set of problems where being right is worth 2–4× the price. Section 8 gives the full decision matrix and the dated watch items (K3 open weights land 27 July 2026 under a Modified MIT license).

2. The landscape

PART 1 · THE BOARD

Six models matter for a quant + AI-engineering practice right now. Two are open-weight Chinese labs (MiniMax, Moonshot), three are US frontier labs (Anthropic ×2, OpenAI), one is Google’s flagship. All six now advertise 1M-token-class context windows — the “who has long context” contest is over; the contest is now price, throughput, and agentic reliability.

Model	Provider	Architecture	Context	Max output	Price in/out (\$/M)	Cache read
MiniMax-M3	MiniMax	428B MoE, ~22B active (128 experts, 4 active) + 600M vision enc.	1M	512K (128K dflt)	\$0.30 / \$1.20	\$0.06
Gemini 3 Pro	Google	undisclosed	1M	64K	\$2 / \$12 (≤200K); \$4/\$18 (>200K)	storage-based
Kimi K3	Moonshot AI	2.8T Stable LatentMoE, 16/896 experts; active params undisclosed (~50–90B bounded est.)	1M	131K dflt → 1M cfg	\$3 / \$15	\$0.30 (mechanics undocumented)
GPT-5.6 Sol	OpenAI	undisclosed	1.05M	128K	\$5 / \$30	\$0.50
Claude Opus 4.8	Anthropic	undisclosed	1M (default)	128K (300K batch beta)	\$5 / \$25; fast \$10/\$50	\$0.50 (min 1,024 tok)
Claude Fable 5	Anthropic	undisclosed	1M	128K	\$10 / \$50	\$1.00

Sources: MiniMax platform pricing + model docs;⁶ Moonshot K3 release + platform docs;⁷ Anthropic Opus 4.8 announcement + pricing + platform docs;⁸ Anthropic Fable page + pricing;⁹ OpenAI pricing;¹⁰ Google Gemini pricing.¹¹ K3 active-parameter bound: §9, gap G1.

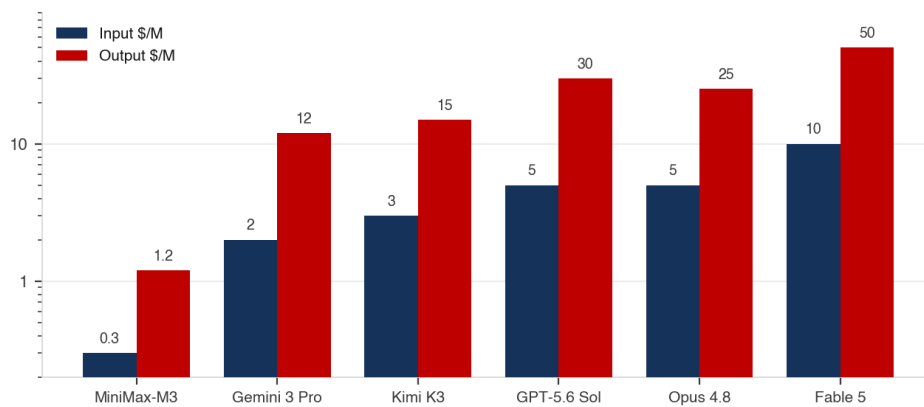


Figure 1. Published API list prices, USD per million tokens (log scale). MiniMax-M3’s \$0.30/\$1.20 is an order of magnitude below every other frontier-class model; Fable 5 anchors the premium end. Fable 5’s new tokenizer (~30% more tokens per document vs older Claudes) effectively multiplies its column by ~1.3.

Three structural notes. First, K3 is the most expensive Chinese-lab model to date — “Sonnet-priced, not cheap-Chinese-priced” — which breaks the assumption that open-weight Chinese labs always undercut on price; M3 carries that flag now.¹² Second, Opus 4.8’s context window is 1M tokens by default (platform documentation) — earlier comparisons citing 200K were working from launch-era material.^[^opusdocs] Third, K3’s active parameter count is officially undisclosed; the circulating “~50B” is naive expert-ratio math ($16/896 \times 2.8T$) that ignores shared and attention parameters. The best-bounded third-party estimate is ~80–90B by K2.6-generation ratio extrapolation — itself speculative. The technical report ships with open weights on **27 July 2026**; that is the resolution date.¹³

3. Metrics for quant research

PART 2 · REASONING UNDER EVIDENCE

Quant research workloads — factor research, mathematical modeling, statistical validation, long-document synthesis — stress four capabilities: graduate-level reasoning, competition math, very long context retrieval, and citation discipline. The published numbers:

Benchmark	MiniMax-M3	Kimi K3	Opus 4.8	Fable 5	GPT-5.6 Sol	Gemini 3 Pro
GPQA-Diamond	92.9 [1-src]	93.5	93.6	92.6	94.1	91.9 (93.8 DT)

Benchmark	MiniMax-M3	Kimi K3	Opus 4.8	Fable 5	GPT-5.6 Sol	Gemini 3 Pro
AIME 2026	N/P ¹	~89.1 (2025) [1-src, unverified]	100 [vendor] / 95.7 [agg.] — conflict	N/P	N/P	MathArena Apex 23.4 (SOTA)
HLE (no tools)	37.1 [1-src]	43.5	49.8 / 45.7 — source conflict	53.3	47.2	37.5 (41.0 DT)
HLE (with tools)	N/P	56.0	57.9	N/P	N/P	N/P
FrontierMath (tier unconfirmed)	N/P	~42 [1-src]	~40 [1-src]	N/P	~39 [1-src]	N/P
FinQA / ConvFinQA	N/P	N/P	N/P	N/P	N/P	N/P
Long-ctx (RULER/NIAH/MRCR-1M)	N/P	N/P	N/P	N/P	N/P	N/P

Sources: Moonshot K3 release table;¹⁴ Anthropic Opus 4.8 system material;¹⁵ evals.report aggregator;¹⁶ Artificial Analysis;¹⁷ Google Gemini 3 Pro model card;¹⁸ shawnhack independent runs [1-src].¹⁹ ¹ M3’s vendor reports IMO-2025/USAMO-2026 above the human gold threshold via a MaxProof test-time-scaffold — a scaffolded result, not raw-model AIME performance.²⁰

Read 1 — the core reasoning band is saturated at the top. GPQA-Diamond now separates six frontier models by 2.2 points. At this compression, GPQA is no longer a selection metric; it is a hygiene check. The discriminating tests have moved to HLE (where Fable 5’s 53.3 still leads by ~4–6 points) and to scaffolded/tool-augmented regimes (HLE-with-tools: Opus 57.9, K3 56.0).

Read 2 — the finance-numeric gap is total and matters. No frontier lab publishes FinQA, ConvFinQA, or any finance-specific numeric-reasoning score for these six models. For a quant practice this means published benchmarks certify *general* reasoning only; finance-numeric reliability must be self-evaluated on proprietary tasks. This brief flags it as the single most important missing metric for the QR track (§9, gap G3).

Read 3 — long-context is now table stakes, but retrieval-quality data lags context-size marketing. All six models advertise ~1M-token windows; none has a current public MRCR-1M or RULER score the others can be compared

against (the leaderboard was unreachable during this research window — \$9, gap G4). K3’s Artificial Analysis long-horizon knowledge-work Elo of 1547 is the best available proxy and places it first among non-Fable models.²¹

Read 4 — math claims need provenance discipline. Anthropic’s vendor-reported AIME 2026 = 100 conflicts with the 95.7 aggregator figure; K3’s “~89.1 AIME 2025” circulates from a single unverified aggregator; FrontierMath figures are column-inferred from one source. All three are labeled here and excluded from the decision matrix. This is what the verification layer is *for*.

4. Metrics for AI engineering & architecture

PART 3 · AGENTIC EVIDENCE

AI-engineering workloads — multi-file code generation, refactors, tool-using agent loops, MCP server work, RAG pipelines — stress a different axis: doing, not knowing. The published numbers:

Benchmark	MiniMax-M3	Kimi K3	Opus 4.8	Fable 5	GPT-5.6 Sol	Gemini 3 Pro
SWE-bench Verified	80.5 [vendor]	N/P	88.6	95.0	N/P	76.2
Terminal-Bench 2.1	66.0 [vendor]	88.3	84.6	84.6	88.8	54.2 on TB 2.0 ²
MCP Atlas	74.2 [vendor]	84.2	N/P	N/P	N/P	N/P
BrowseComp	83.5 [vendor]	91.2	79.3 (Opus 4.7; 4.8 N/P)	N/P	92.2	N/P
DeepSWE	N/P	67.5	59.0	70.0	73.0	N/P
τ ² -bench Telecom	N/P	N/P	94.4	N/P	N/P	N/P
OSWorld	N/P	N/P	83.4 (v1) ³	N/P	62.6 (v2.0) ³	N/P
M3-only agentic suite	Claw-Eval 74.5 (#1); PostTrainBench 37.1 (#3); SkillsBench 53; SWE-bench Pro 59.0	—	—	—	—	—

Sources: MiniMax M3 release blog + HF model card;²² Moonshot K3 release table;²³ Anthropic;²⁴ OpenAI;²⁵ evals.report;²⁶ Artificial Analysis.²⁷ ² Gemini’s 54.2 is on Terminal-Bench 2.0 — do not cross-compare versions. ³ OSWorld v1 and v2.0 scores are not comparable across versions.

Read 1 — K3’s agentic debut is real, with a vendor asterisk. Terminal-Bench 88.3 and MCP Atlas 84.2 are the strongest release-day agentic numbers any open-weight model has posted. Every K3 figure in this table is vendor-published on Moonshot’s own harness; none is independently replicated as of 18 July. The correct confidence level is *high that the vendor ran them, medium on the numbers* — independent reruns are the watch item (§8).

Read 2 — M3’s profile is “fast generalist with real agentic chops.” A vendor SWE-bench of 80.5 (unreplicated) plus the top Claw-Eval score plus the fastest measured throughput in the field makes M3 the natural bulk-workhorse: good enough to trust with high-frequency coding tasks, cheap enough to run at volume, fast enough to keep feedback loops tight.

Read 3 — SWE-bench Verified is the metric K3 won’t fight on. Moonshot publishes DeepSWE, FrontierSWE (81.2), and SWE Marathon (42.0) instead. Treat any “K3 SWE-bench 76.8%” figure as non-credible — no official score exists (§9, D1). Until an independent rerun appears, the defensible coding statement is: Fable 5 > Opus 4.8 > {K3, M3} on real-issue resolution, with K3 strongest on terminal-style execution.

Read 4 — the toolchain layer still belongs to Anthropic. τ^2 -bench Telecom 94.4, the Claude Code harness, mature MCP/hook ecosystems, and mid-conversation system messages (which preserve prompt cache across agent-loop turns) are production features no benchmark table captures. For architecture work where the model must live *inside* a tool-using system, this ecosystem maturity is itself a metric — currently unquantified but decisive (§9, gap G5).

5. Operational metrics

PART 4 · THE PHYSICS

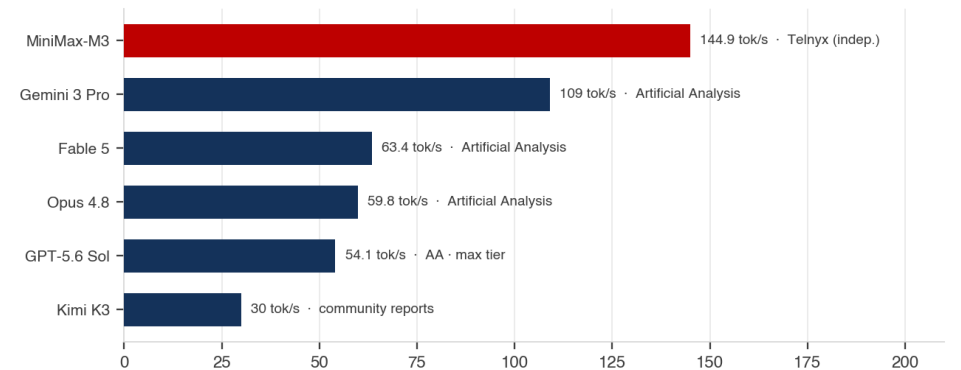


Figure 2. Measured output throughput, tokens/second. MiniMax-M3 leads by ~2.3× over the next fastest (Telnyx and Together, independent measurements); Kimi K3’s ~30 t/s rests on first-week community reports. Sources: Telnyx/Together benchmarks;²⁸ Artificial Analysis;²⁹ r/LocalLLaMA.³⁰

Metric	MiniMax-M3	Kimi K3	Opus 4.8	Fable 5	GPT-5.6 Sol	Gemini 3 Pro
Throughput (tok/s)	144.9 / 119.8 (indep.)	~30 (community)	59.8	63.4	54.1 (max tier)	~109
TTFT (p50)	N/P	N/P	1.9s	1.4s	8.2s (high)	32.7s first-answer
TTFT at max reasoning	—	always-on	60.9s	109s	145.6s	—
Context window	1M	1M	1M	1M	1.05M	1M
Max output	512K (128K dflt)	131K → 1M cfg	128K (300K batch beta)	128K	128K	64K
Uptime / SLA	N/P	N/P	N/P	N/P	N/P	N/P

Sources: provider documentation as cited in §2; Artificial Analysis speed/TTFT measurements;³¹ Contra Collective decode-rate tests.³²

The operational story is M3’s second surprise. A model at \$0.30/\$1.20 that also runs 2.3–5× faster than everything else measured is not a compromise — for latency-sensitive, high-frequency workloads (interactive coding loops, rapid factor-iteration, live data munging) it is the *performance* choice on two axes at once. K3’s ~30 t/s is its most tangible daily friction: long synthesis jobs are measured in tens of minutes, and its max-only reasoning tier means there is no cheaper, faster mode to drop into. The max-reasoning TTFT figures (61–146 seconds across labs) quantify why “always-on max” is a workflow decision, not just a price decision.

6. Cost-per-task economics

PART 5 · THE ARITHMETIC THAT DECIDES ROUTING

List prices are inputs; task economics are what a practice actually pays. Using the verified prices from §2 ($\text{cost} = \text{in} \times P_{\text{in}} / 1\text{M} + \text{out} \times P_{\text{out}} / 1\text{M}$, no caching), across four workload shapes representative of quant + AI-engineering work:

Workload shape (tokens in/out)	MiniMax- M3	Gemini 3 Pro	Kimi K3	GPT-5.6 Sol	Opus 4.8	Fable 5
Backtest codegen — 15K / 6K	\$0.012	\$0.102	\$0.135	\$0.255	\$0.225	\$0.450
Research synthesis — 80K / 15K	\$0.042	\$0.340	\$0.465	\$0.850	\$0.775	\$1.550
Agentic refactor — 120K / 25K	\$0.066	\$0.540	\$0.735	\$1.350	\$1.225	\$2.450
Routine edits — 4K / 1K	\$0.002	\$0.020	\$0.027	\$0.050	\$0.045	\$0.090

Arithmetic shown in full in §9 note N1; prices verified in §2. Fable 5’s tokenizer (~+30% tokens/document) effectively multiplies its column by ~1.3. No caching, batch, or subscription effects included.

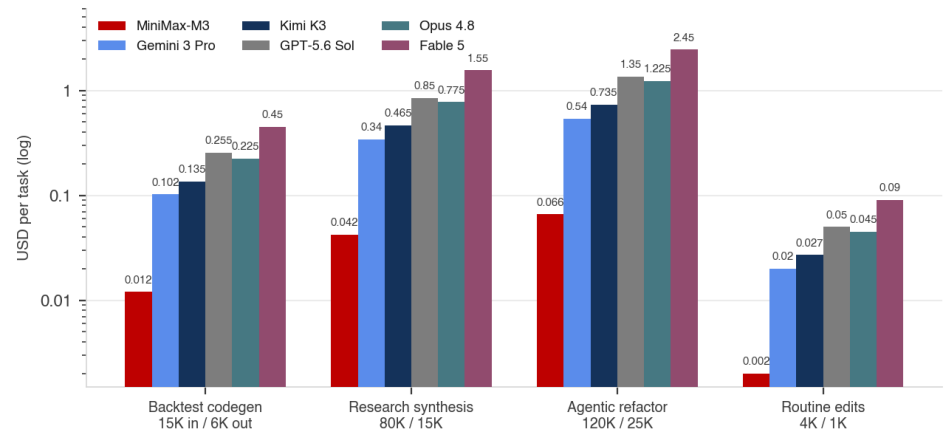


Figure 3. USD per task by workload shape (log scale). M3’s advantage is structural (10–20×), not marginal; the middle tier (Gemini–K3–Opus) clusters within ~2×; Fable 5 prices as the premium tier it benchmarks as.

The scale intuition: a practice running 50M input + 10M output tokens monthly pays, at list, **\$27/month on M3, \$220 on Gemini 3 Pro (≤200K tier), \$300 on K3, \$500 on Opus 4.8, \$550 on GPT-5.6 Sol, and \$1,000 on Fable 5**. Over a quarter that is the difference between a rounding error and a line item — and it reframes “which model is default” as a budget-architecture question with a capability overlay, not a benchmark-leaderboard question.

Cache economics: where Opus claws the middle back. Opus 4.8’s 1,024-token cache minimum brings most real system prompts into the 90%-discount regime. Effective input price at cache-hit fraction f is $5(1-f) + 0.50f$; setting equal to K3’s \$3 base gives $f \approx 0.44$. Agentic loops routinely run 70–90% repeated prefix — deep into the region where cached Opus input undercuts K3. K3 lists a \$0.30 cache-read price but publishes no minimum, TTL, or invalidation behavior, and lacks mid-conversation system-message support; treat K3 cache as a bonus if it hits, never as a planning assumption.

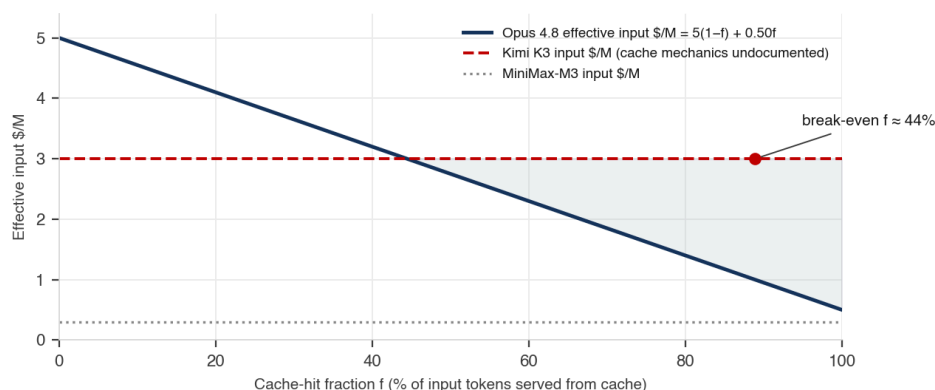


Figure 4. Opus 4.8 effective input price vs cache-hit fraction. Above $f \approx 44\%$, Opus input undercuts K3’s base input price; the shaded region is where agentic workloads actually live. M3’s \$0.30 floor is shown for reference — no cache mechanism beats it on price.

The verbosity tax is the counterweight to K3’s per-token price. K3 runs one reasoning tier — max — and it shows: 16,658 output tokens (13,241 reasoning) from a 95-token prompt in one independent test (\$0.25 for a trivial query), and ~130M tokens burned in its Artificial Analysis evaluation run ($\approx 2\times$ average verbosity).³³ Budget K3 by *output tokens*, not by request count; on poorly-scoped prompts its effective cost can double its table price. M3’s MSA attention architecture (vendor-claimed $9\times$ prefill / $15\times$ decode efficiency vs its predecessor at 1M context) is the architectural reason its price-performance holds at long context.³⁴

7. The macro frame: why the price structure looks like this

PART 6 · CONTEXT FROM THE FUNDING AND COST LITERATURE

The price table in §2 is not arbitrary — it sits on top of two well-documented structural facts about the 2026 AI economy, drawn from companion cost-economics and funding research (data as of June 2026).³⁵

Fact 1 — the two superpowers fund AI in opposite directions, and it shows in model pricing. US private AI investment ran ~\$285.9B in 2025 against China’s \$12.4B (a $23\times$ private-capital ratio, Stanford HAI) — yet China

sits within roughly half a year of the frontier, because its labs compete on *efficiency* rather than capital.³⁶ The US model is private-capital-first (mega-rounds plus hyperscaler capex of \$379B in FY2025, guided toward ~\$760B in FY2026); the China model is state-capital-first (a ~\$138B guidance-fund target plus national-champion capex) with **open-weight, price-aggressive models as the strategic wedge** — commoditizing exactly what US labs monetize.³⁷ K3 at \$3/\$15 and M3 at \$0.30/\$1.20 are this strategy expressed as price lists.

Fact 2 — training cost is a layering problem, and sparse-MoE reuse is China’s hedge against export controls. Public “training cost” figures conflate four layers: final-run compute (DeepSeek-V3’s famous ~\$5.6M), compute + ablations (~\$6–10M), all-in program (~\$15–35M), and company-wide strategic spend (\$100M–\$1B+).³⁸ Most Chinese flagships train for **tens of millions, not billions**, via four reuse techniques — checkpoint inheritance, distillation, continued pretraining, and sparse MoE with low active-parameter counts (DeepSeek-V3: 671B total / 37B active; MiniMax-01: 456B / 45.9B).³⁹ K3 (2.8T, 16/896 experts) and M3 (428B, ~22B active) are direct descendants of this design lineage — large-model quality at small-model serving compute. Chips and staff are ~68% of a frontier run’s cost while energy is 2–6%: the cost contest was never an electricity story.⁴⁰

Why this matters for a buyer. A \$100M training run amortized over 100 trillion served tokens adds \$1 per million tokens — below every list price in §2. The strategic implication: **API prices for open-weight-lineage models have structural room to fall further**, while premium US flagships must justify themselves on capability deltas and ecosystem, not cost. Lock no long-term spend to today’s prices; revisit the routing table quarterly. The same literature carries a caution worth repeating: ~95% of enterprise generative-AI pilots report no measurable ROI (MIT) against a ~6× capex-to-realized-revenue gap — model selection discipline (this brief’s purpose) is how a practice stays in the 5%.⁴¹

8. Decision framework

PART 7 · THE ROUTING TABLE

Three-tier default policy for a quant + AI-engineering practice:

Workload class	Examples	Route to	Why
Bulk / high-frequency	Data plumbing, routine codegen, file transforms, first drafts, test scaffolding	MiniMax-M3	Market-floor price + fastest measured throughput; 80.5 vendor SWE-bench is ample at this tier

Workload class	Examples	Route to	Why
Long-horizon synthesis	Multi-source research briefs, literature sweeps, long-document QA	Kimi K3	Best-in-class long-horizon Elo; 1M context; -21% output-token efficiency vs predecessor
Terminal-agentic execution	CLI-driven tasks, shell-heavy automation	Kimi K3	Terminal-Bench 88.3, MCP Atlas 84.2 (vendor-pending replication)
Cache-friendly agent loops	Multi-turn tool-using systems with large static system prompts	Opus 4.8	f≈44% break-even (\$6); mature MCP/hooks ecosystem; mid-conv system messages
Precision-critical writing	Client-facing analysis, anything where one wrong number costs trust	Opus 4.8	Self-reported lowest incorrect-rate of six models tested; strong instruction-following
Hardest-reasoning reserve	Novel proof-shaped problems, architecture decisions you can't afford to get wrong	Fable 5	HLE 53.3, SWE-bench 95.0 — buy the top tier only where its delta is monetizable
Max-throughput long context	1M-token ingest where speed matters more than peak quality	Gemini 3 Pro	~109 t/s + 1M context + \$2/\$12 entry price

If forced to a single default: MiniMax-M3 for cost-driven practices, Kimi K3 for synthesis-driven practices, Opus 4.8 for agent-systems-driven practices. The evidence supports all three — which is the point of routing.

Sensitivity — what would change this table (dated watch items):

- 27 July 2026 — K3 open weights + technical report (Modified MIT).**
Resolves the active-parameter null (\$9 G1); unlocks self-host economics (break-even historically ~50–100M output tokens/month — far above a solo practice's volume, so the API decision stands, but vendor-neutral serving via third-party hosts could cut K3's effective price).
- Independent replication of K3's agentic table** (Terminal-Bench, MCP Atlas). Upgrades confidence from vendor-medium to high — or revises \$4 Read 1 downward.

3. **K3 cache-mechanics documentation.** If Moonshot publishes a real cache floor + TTL, the §6 break-even analysis needs a second curve.
4. **M3 third-party SWE-bench replication.** M3’s 80.5 is vendor-run; an independent confirmation would make §8’s bulk tier bulletproof.
5. **Any published FinQA-class or MRCR-1M result.** First mover in either gap materially shifts the QR recommendation.

9. Method, verification & gap log

APPENDIX · PROVENANCE

Method. This brief was produced by a multi-agent research pipeline: a research agent gathered and cross-checked primary sources (every headline number required two independent citations or an explicit [1-src] label); a cost-modeling agent computed task economics from verified prices; two independent verification gates then audited the draft — one for arithmetic/consistency/claim-labeling (six findings, all applied), one for disclosure hygiene. Figures were generated from the same verified tables. Nothing in this brief is unsourced; where the evidence does not exist, the gap is stated rather than filled.

Honest nulls (declared, not filled):

#	Gap	Status	Resolution path
G1	K3 active parameter count	Officially undisclosed; bounded ~50–90B (methodology-labeled)	Technical report, 27 Jul 2026
G2	K3 SWE-bench Verified	Not published; vendor uses DeepSWE/FrontierSWE/SWE Marathon	Independent rerun
G3	Finance-numeric benchmarks (FinQA/ConvFinQA)	Not published for any of the six models	Self-evaluation on proprietary tasks
G4	Long-context retrieval (MRCR-1M/RULER)	Leaderboard unreachable in research window; no cross-model current data	Re-fetch at next refresh
G5	Agentic-ecosystem maturity metric	No benchmark captures MCP/hooks/tooling depth	Qualitative assessment only
G6	Uptime/SLA comparisons	No public breach or SLA data for any model	Operational monitoring
G7	M3 GPQA/HLE/MMLU-Pro/LiveCodeBench	Single-source independent [1-src]	Await corroboration

Do-not-use register (claims in circulation that fail verification): D1 — any “K3 SWE-bench 76.8%” (never official). D2 — “K3 ~50B active params” stated as fact (naive ratio math). D3 — K3 “AIME 89.1%” (single unverified aggregator). D4 — FrontierMath figures (attribution inferred). D5 — cross-version OSWorld or Terminal-Bench 2.0-vs-2.1 comparisons. D6 — Opus 4.8 “AA Index 61.4” (stale launch version; current 56). D7 — Fable 5 “\$8/\$40” pricing (verified \$10/\$50). D8 — “Opus 4.8 200K context” (1M default per current platform docs). D9 — Gemini “2M context” (official 1M).

Note N1 — cost arithmetic. $\text{cost} = \text{in} \times P_{\text{in}} / 1\text{M} + \text{out} \times P_{\text{out}} / 1\text{M}$. Example (agentic refactor, 120K/25K): M3 = $0.12 \times 0.30 + 0.025 \times 1.20 = \$0.036 + \$0.030 = \text{\$0.066}$; K3 = $0.12 \times 3 + 0.025 \times 15 = \$0.36 + \$0.375 = \text{\$0.735}$; Opus = $0.12 \times 5 + 0.025 \times 25 = \$0.60 + \$0.625 = \text{\$1.225}$; Fable = $\$1.20 + \$1.25 = \text{\$2.450}$; Sol = $\$0.60 + \$0.75 = \text{\$1.350}$; Gemini = $\$0.24 + \$0.30 = \text{\$0.540}$. All \$6 figures derive identically and were re-verified by the quality gate.

Sources

Prepared by STELLA Research · Multi-agent pipeline: research → cost modeling → quality gate → disclosure gate → render · Generated 18 July 2026 · This brief may be shared freely with attribution.

-
1. Moonshot AI, “Kimi K3” release blog + platform documentation (16–17 Jul 2026) — architecture, benchmark table, context window, sampling parameters, open-weights commitment. <https://www.kimi.com/blog/kimi-k3>↩
 2. MiniMax platform pricing documentation + MiniMax-M3 model page; corroborated by Runware and Vercel AI Gateway model listings. <https://platform.minimax.io/docs/guides/pricing-paygo>↩
 3. MiniMax, “MiniMax-M3” release blog + Hugging Face model card — SWE-bench Verified 80.5, Terminal-Bench 66.0, MCP Atlas 74.2, Claw-Eval 74.5, architecture (428B/22B-active, MSA attention); NVIDIA developer blog on MSA efficiency. <https://www.minimax.io/blog/minimax-m3>↩
 4. Anthropic, “Claude Opus 4.8” announcement, pricing page, and platform documentation (1M default context, 1,024-token cache minimum, mid-conversation system messages, fast mode \$10/\$50); Simon Willison’s Opus 4.8 review (28 May 2026). <https://www.anthropic.com/news/claude-opus-4-8>↩
 5. Simon Willison, “Kimi K3” deep-dive (16 Jul 2026) — 95-token prompt → 16,658 output tokens = \$0.25; tokenizer anomaly; per-task cost \$0.94. <https://simonwillison.net/2026/Jul/16/kimi-k3/>↩
 6. MiniMax platform pricing documentation + MiniMax-M3 model page; corroborated by Runware and Vercel AI Gateway model listings. <https://platform.minimax.io/docs/guides/pricing-paygo>↩
 7. Moonshot AI, “Kimi K3” release blog + platform documentation (16–17 Jul 2026) — architecture, benchmark table, context window, sampling parameters, open-weights commitment. <https://www.kimi.com/blog/kimi-k3>↩

8. Anthropic, “Claude Opus 4.8” announcement, pricing page, and platform documentation (1M default context, 1,024-token cache minimum, mid-conversation system messages, fast mode \$10/\$50); Simon Willison’s Opus 4.8 review (28 May 2026).
<https://www.anthropic.com/news/claude-opus-4-8>↵
9. Anthropic, Claude Fable page + platform pricing (\$10/\$50; cache write \$12.50/\$20, hits \$1; tokenizer note). <https://www.anthropic.com/claude/fable>↵
10. OpenAI, API pricing documentation (GPT-5.6 Sol \$5/\$30, cached input \$0.50).
<https://openai.com/api/pricing>↵
11. Google, Gemini API pricing (Gemini 3 Pro \$2/\$12 ≤200K, \$4/\$18 >200K).
<https://ai.google.dev/pricing>↵
12. The Decoder — “K3 is Sonnet-priced”: most expensive Chinese-lab model to date. <https://the-decoder.com>↵
13. Tosea.ai, “Kimi K3 complete guide” (undisclosed active params; ~80–90B K2.6-ratio extrapolation, labeled speculation); Latent.Space coverage (“2.8T-A50B” naming); dev.to/tokenmixai ratio-critique. <https://tosea.ai/blog/kimi-k3-complete-guide>↵
14. Moonshot AI, “Kimi K3” release blog + platform documentation (16–17 Jul 2026) — architecture, benchmark table, context window, sampling parameters, open-weights commitment.
<https://www.kimi.com/blog/kimi-k3>↵
15. Anthropic, “Claude Opus 4.8” announcement, pricing page, and platform documentation (1M default context, 1,024-token cache minimum, mid-conversation system messages, fast mode \$10/\$50); Simon Willison’s Opus 4.8 review (28 May 2026).
<https://www.anthropic.com/news/claude-opus-4-8>↵
16. evals.report benchmark aggregator — Opus 4.8 SWE-bench 88.6, HLE 49.8, τ^2 -Telecom 94.4, ARC-AGI-2 72.08, OSWorld v1 83.4. <https://evals.report>↵
17. Artificial Analysis — Intelligence Index v4.1 (Fable 60, K3 57, Opus 56, M3 ~55), speed/TTFT measurements, Terminal-Bench Hard leaderboard. <https://artificialanalysis.ai>↵
18. Google, Gemini 3 Pro model card (GPQA-D 91.9/93.8 DT, HLE 37.5/41.0 DT, SWE-bench 76.2, MathArena Apex 23.4). <https://deepmind.google>↵
19. shawnhack.com independent model runs [single-source]: M3 GPQA 92.9, HLE 37.1, MMLU-Pro 84.2, LiveCodeBench 82.2; corroborated directionally by Kili Technology data story (AA ~55).
<https://shawnhack.com>↵
20. MiniMax MaxProof blog — IMO-2025/USAMO-2026 above human gold threshold under test-time scaffolding (scaffolded result, not raw-model AIME). <https://www.minimax.io/blog>↵
21. Artificial Analysis long-horizon knowledge-work Elo — K3 1547, first among non-Fable models (as of 17 Jul 2026). <https://artificialanalysis.ai>↵
22. MiniMax, “MiniMax-M3” release blog + Hugging Face model card — SWE-bench Verified 80.5, Terminal-Bench 66.0, MCP Atlas 74.2, Claw-Eval 74.5, architecture (428B/22B-active, MSA attention); NVIDIA developer blog on MSA efficiency. <https://www.minimax.io/blog/minimax-m3>↵
23. Moonshot AI, “Kimi K3” release blog + platform documentation (16–17 Jul 2026) — architecture, benchmark table, context window, sampling parameters, open-weights commitment.
<https://www.kimi.com/blog/kimi-k3>↵
24. Anthropic, “Claude Opus 4.8” announcement, pricing page, and platform documentation (1M default context, 1,024-token cache minimum, mid-conversation system messages, fast mode \$10/\$50); Simon Willison’s Opus 4.8 review (28 May 2026).
<https://www.anthropic.com/news/claude-opus-4-8>↵

25. OpenAI, API pricing documentation (GPT-5.6 Sol \$5/\$30, cached input \$0.50).
<https://openai.com/api/pricing>↵
26. evals.report benchmark aggregator — Opus 4.8 SWE-bench 88.6, HLE 49.8, τ^2 -Telecom 94.4, ARC-AGI-2 72.08, OSWorld v1 83.4. <https://evals.report>↵
27. Artificial Analysis — Intelligence Index v4.1 (Fable 60, K3 57, Opus 56, M3 ~55), speed/TTFT measurements, Terminal-Bench Hard leaderboard. <https://artificialanalysis.ai>↵
28. Telnix and Together independent throughput benchmarks for MiniMax-M3 (144.9 and 119.8 tok/s). <https://www.telnix.com>↵
29. Artificial Analysis — Intelligence Index v4.1 (Fable 60, K3 57, Opus 56, M3 ~55), speed/TTFT measurements, Terminal-Bench Hard leaderboard. <https://artificialanalysis.ai>↵
30. r/LocalLLaMA first-week K3 hands-on reports (~30 t/s; 35-minute frontend task; quality assessments). <https://www.reddit.com/r/LocalLLaMA>↵
31. Artificial Analysis — Intelligence Index v4.1 (Fable 60, K3 57, Opus 56, M3 ~55), speed/TTFT measurements, Terminal-Bench Hard leaderboard. <https://artificialanalysis.ai>↵
32. Contra Collective decode-rate tests (Opus 4.8 ~52 t/s, Fable 5 ~71 t/s, TTFT p50 figures).
<https://contracollective.com>↵
33. Simon Willison, “Kimi K3” deep-dive (16 Jul 2026) — 95-token prompt → 16,658 output tokens = \$0.25; tokenizer anomaly; per-task cost \$0.94. <https://simonwillison.net/2026/Jul/16/kimi-k3/>↵
34. MiniMax, “MiniMax-M3” release blog + Hugging Face model card — SWE-bench Verified 80.5, Terminal-Bench 66.0, MCP Atlas 74.2, Claw-Eval 74.5, architecture (428B/22B-active, MSA attention); NVIDIA developer blog on MSA efficiency. <https://www.minimax.io/blog/minimax-m3>↵
35. “AI Workload Cost: The United States vs. China” and “AI Funding & Economics: The United States vs. China” — companion deep-research briefings (data as of 26 June 2026): four-layer training-cost framework, node-hour TCO model, reuse techniques, amortization math, MIT 95%-pilots statistic, capex/revenue gap. Sources therein: EIA, LBNL, Uptime Institute, Epoch AI, DeepSeek-V3 technical report, company filings via Perplexity Finance.↵
36. Stanford HAI, AI Index 2026 — US \$285.9B vs China \$12.4B private AI investment (2025), like-for-like private-only basis. <https://hai.stanford.edu/ai-index>↵
37. “AI Workload Cost: The United States vs. China” and “AI Funding & Economics: The United States vs. China” — companion deep-research briefings (data as of 26 June 2026): four-layer training-cost framework, node-hour TCO model, reuse techniques, amortization math, MIT 95%-pilots statistic, capex/revenue gap. Sources therein: EIA, LBNL, Uptime Institute, Epoch AI, DeepSeek-V3 technical report, company filings via Perplexity Finance.↵
38. DeepSeek-V3 Technical Report — 2.788M H800 GPU-hours, ~\$5.576M direct run cost, 671B/37B-active MoE. <https://arxiv.org/abs/2412.19437>↵
39. “AI Workload Cost: The United States vs. China” and “AI Funding & Economics: The United States vs. China” — companion deep-research briefings (data as of 26 June 2026): four-layer training-cost framework, node-hour TCO model, reuse techniques, amortization math, MIT 95%-pilots statistic, capex/revenue gap. Sources therein: EIA, LBNL, Uptime Institute, Epoch AI, DeepSeek-V3 technical report, company filings via Perplexity Finance.↵
40. Epoch AI, “The rising costs of training frontier AI models” — chips and staff dominate; energy ~2–6%. <https://arxiv.org/abs/2405.21015>↵

41. “AI Workload Cost: The United States vs. China” and “AI Funding & Economics: The United States vs. China” — companion deep-research briefings (data as of 26 June 2026): four-layer training-cost framework, node-hour TCO model, reuse techniques, amortization math, MIT 95%-pilots statistic, capex/revenue gap. Sources therein: EIA, LBNL, Uptime Institute, Epoch AI, DeepSeek-V3 technical report, company filings via Perplexity Finance. ↩